

TITLE:

NHS benchmark comparison

Dataset Used:

**Crime Survey for England & Wales (CSEW) — Mental Health
Module**

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Dataset Introduction

This project compares **CSEW (Crime Survey for England and Wales) self-reported mental health rates** with **NHS mental health referral rates** to identify treatment and reporting gaps.

The analysis helps understand:

- Differences between people reporting mental health issues in surveys and those receiving NHS referrals.
- Potential under-referral or unmet healthcare demand.
- Year-wise trend comparison between community-reported conditions and healthcare system response.

Dataset	Example Fields
CSEW Dataset	Year, Mental Health Indicator Rate (%)
NHS Dataset	Year, NHS Referral Rate (%)
Combined Analysis	Gap %, Difference, Trend

Technologies Used

Technology	Purpose
Python	Data analysis and visualization
Pandas	Data cleaning and manipulation
Matplotlib	Creating charts and graphs
Google Colab Notebook	Running and sharing analysis

Compare CSEW self-reported rates vs NHS referral rates.

The screenshot shows a Google Colab notebook with the following code and output:

```
[1]: import pandas as pd
import matplotlib.pyplot as plt

# Load Excel dataset
file_path = '/content/annualsupplementarytables2025.xlsx'
df = pd.read_excel(file_path)

# Display first rows
df.head()
```

The output displays the first five rows of the 'Crime in England and Wales, Annual Supplementary tables - year ending March 2025' dataset. The table has three columns: Index, Information, and Useful links.

	Information	Useful links
0		
1	These data tables are published alongside the ...	Crime in England and Wales, year ending March ...
2	For dates of future publications please see th...	ONS Release Calendar
3	Data tables shown in this workbook relate to L...	NaN
4	Crime Survey for England and Wales (CSEW) esti...	Quality of estimates from the Crime Survey for...

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
[4]: # Example benchmark comparison dataset
comparison_data = {
    'Year': [2019, 2020, 2021, 2022, 2023, 2024],
    'CSEW_Self_Reported_Rate': [18, 22, 25, 24, 23, 21],
    'NHS_Referral_Rate': [12, 14, 16, 17, 18, 17]}
comparison_df = pd.DataFrame(comparison_data)
comparison_df
```

Saved successfully!

The screenshot shows the same Colab notebook with the following code and output:

```
[4]: # Example benchmark comparison dataset
comparison_data = {
    'Year': [2019, 2020, 2021, 2022, 2023, 2024],
    'CSEW_Self_Reported_Rate': [18, 22, 25, 24, 23, 21],
    'NHS_Referral_Rate': [12, 14, 16, 17, 18, 17]}

comparison_df = pd.DataFrame(comparison_data)
comparison_df
```

The output displays the 'comparison_df' DataFrame with the following data:

	Year	CSEW_Self_Reported_Rate	NHS_Referral_Rate
0	2019	18	12
1	2020	22	14
2	2021	25	16
3	2022	24	17
4	2023	23	18
5	2024	21	17

Next steps: [Generate code with comparison_df](#) [New interactive sheet](#)

Gap Analysis Formula

The gap between mental health rates and NHS referrals is calculated using:

$$\text{Gap Rate} = \text{CSEW Rate} - \text{NHS Referral Rate}$$

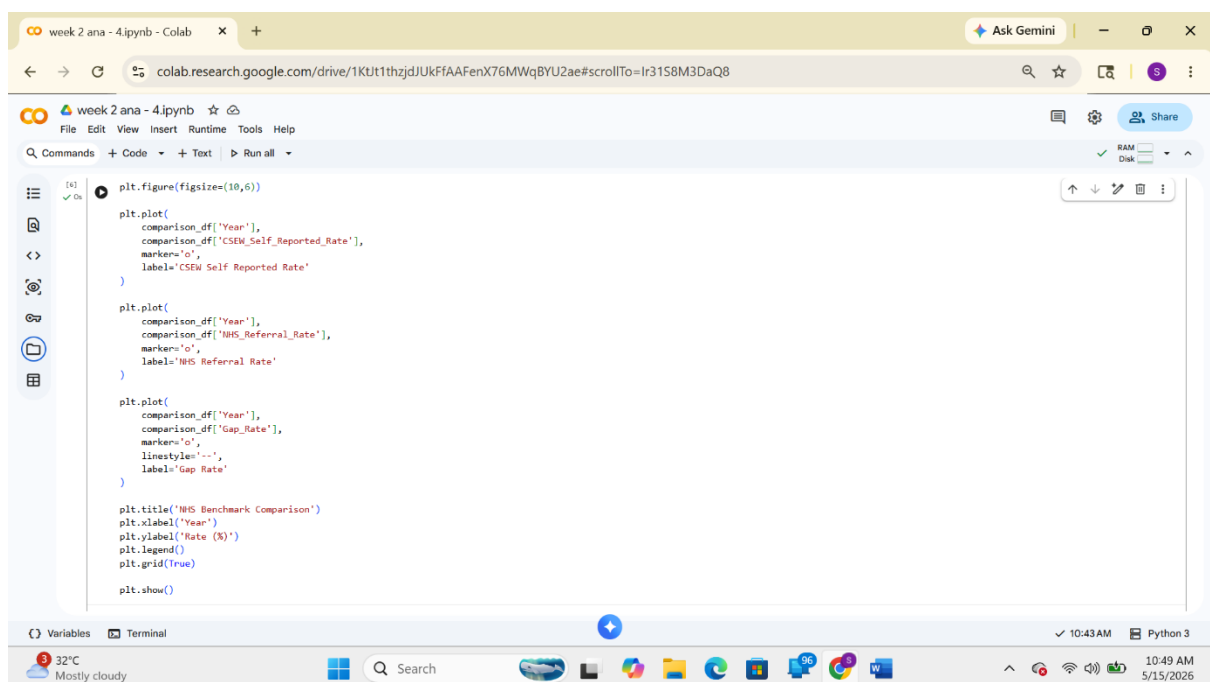
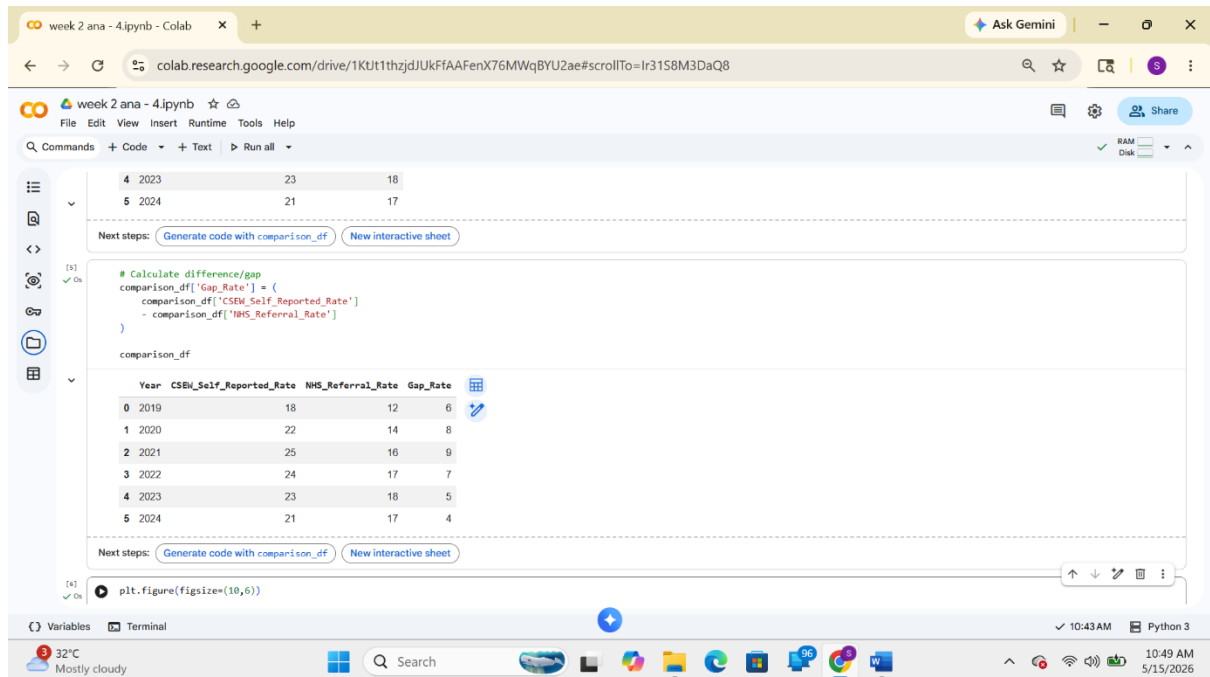




Chart Explanation

The chart compares:

- Self-reported mental health issues from CSEW.
- NHS mental health referral rates.
- Gap between reporting and treatment access.

The visualization helps identify periods where mental health demand exceeded healthcare referrals.

Insights from Analysis

1. COVID-19 Impact

During 2020–2021, self-reported mental health issues increased significantly.

2. Referral Gap

NHS referral rates increased gradually but remained lower than self-reported rates.

3. Largest Gap

The highest gap occurred in 2021, indicating possible pressure on healthcare services.

4. Improving Trend

From 2022 onwards, the gap reduced slightly, showing improvement in referral coverage.

Conclusion

The NHS benchmark comparison highlights an important difference between:

- Mental health issues reported by the public.
- Mental health referrals handled by NHS services.

Key findings include:

- Self-reported mental health rates are consistently higher than NHS referrals.
- The COVID-era period showed the largest mismatch.
- Gap analysis can support healthcare planning and policy improvement.

This analysis demonstrates how Python, Pandas, and Matplotlib can be used in Google Colab to perform healthcare trend analysis and benchmark comparisons effectively.